Numerical Method Lab Report Manual



ગતિ શક્તિ વિશ્વવિદ્યાલય
गति शक्ति विश्वविद्यालय
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```
%{
EXP 1
Operations on three variables (a, b, c)
Add, Subtract, Multiply, Divide, Square, Square-root,
Remainder, Average
%}

a = 10;
b = 15;
c = 20;

% Addition
add_result = a + b + c;

% Subtraction
sub_result = a - b - c;

% Multiplication
mult_result = a * b * c;

% Division
div_result = a / b / c;

% Squares
square_a = a^2;
square_b = b^2;
square_c = c^2;

% Square roots
sqrt_a = sqrt(a);
sqrt_b = sqrt(b);
sqrt_c = sqrt(c);

% Remainder (modulus)
rem_ab = mod(a, b);   % remainder of a ÷ b
rem_bc = mod(b, c);   % remainder of b ÷ c
rem_ca = mod(c, a);   % remainder of c ÷ a

% Average
avg_result = (a + b + c) / 3;

% Display results
disp('Addition:'), disp(add_result)
disp('Subtraction:'), disp(sub_result)
disp('Multiplication:'), disp(mult_result)
disp('Division:'), disp(div_result)
disp('Squares:'), disp([square_a square_b square_c])
disp('Square roots:'), disp([sqrt_a sqrt_b sqrt_c])
disp('Remainders [a mod b, b mod c, c mod a]:'), disp([rem_ab rem_bc rem_ca])
disp('Average:'), disp(avg_result)
```

```
Addition:
    45
```

```
Subtraction:
   -25

Multiplication:
        3000

Division:
    0.0333

Squares:
   100   225   400

Square roots:
    3.1623    3.8730    4.4721

Remainders [a mod b, b mod c, c mod a]:
    10    15     0

Average:
    15
```

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```
%{
EXP 2
Matrix operations with A and B
%}

% Define 4x3 matrices
A = [1 2 3; 4 5 6; 7 8 9; 1 0 1];
B = [0 1 0; 1 0 1; 2 2 2; 3 3 3];

% Combine matrices
C_h = [A B];   % Horizontal combine (4x6)
C_v = [A; B];  % Vertical combine (8x3)

% Rank
rankA = rank(A);

% Trace (only for square submatrix)
traceA = trace(A(1:3,1:3));

% Flip
A_lr = fliplr(A); % Left-right flip
A_ud = flipud(A); % Up-down flip

% Modify diagonal elements (+10)
A_diagmod = A;
for i = 1:min(size(A))
    A_diagmod(i,i) = A_diagmod(i,i) + 10;
end

% Add / Subtract matrices
AddAB = A + B;
SubAB = A - B;

% Transpose
A_T = A';

% Inverse of a 3x3 matrix
M = [2 1 1; 1 2 1; 1 1 2];
invM1 = inv(M);               % Method 1
invM2 = adjoint(M)/det(M);    % Method 2 (using custom adjoint function)

% Display results
disp('Horizontal combination C_h ='); disp(C_h);
disp('Vertical combination C_v ='); disp(C_v);
disp(['Rank of A = ', num2str(rankA)]);
disp(['Trace of 3x3 submatrix of A = ', num2str(traceA)]);
disp('A left-right flip ='); disp(A_lr);
disp('A up-down flip ='); disp(A_ud);
disp('A with diagonal modified (+10) ='); disp(A_diagmod);
disp('A + B ='); disp(AddAB);
disp('A - B ='); disp(SubAB);
disp('Transpose of A ='); disp(A_T);
```

```
disp('Inverse of M (method 1) ='); disp(invM1);
disp('Inverse of M (method 2 using adjoint) ='); disp(invM2);
```

```
Horizontal combination C_h =
     1     2     3     0     1     0
     4     5     6     1     0     1
     7     8     9     2     2     2
     1     0     1     3     3     3

Vertical combination C_v =
     1     2     3
     4     5     6
     7     8     9
     1     0     1
     0     1     0
     1     0     1
     2     2     2
     3     3     3

Rank of A = 3
Trace of 3x3 submatrix of A = 15
A left-right flip =
     3     2     1
     6     5     4
     9     8     7
     1     0     1

A up-down flip =
     1     0     1
     7     8     9
     4     5     6
     1     2     3

A with diagonal modified (+10) =
    11     2     3
     4    15     6
     7     8    19
     1     0     1

A + B =
     1     3     3
     5     5     7
     9    10    11
     4     3     4

A - B =
     1     1     3
     3     5     5
     5     6     7
    -2    -3    -2

Transpose of A =
     1     4     7     1
     2     5     8     0
```

```
     3     6     9     1

Inverse of M (method 1) =
    0.7500   -0.2500   -0.2500
   -0.2500    0.7500   -0.2500
   -0.2500   -0.2500    0.7500

Inverse of M (method 2 using adjoint) =
    0.7500   -0.2500   -0.2500
   -0.2500    0.7500   -0.2500
   -0.2500   -0.2500    0.7500
```

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```
%{
Exp 3.1
To find root of quadratic
equation x^2 + 3*x - 4 = 0
%}

a = 1; b = 3; c = -4; % Coefficients of the quadratic equation
d = b^2 - 4*a*c; % Calculate the discriminant

root1 = (-b+sqrt(d))/(2*a); %find root 1
root2 = (-b-sqrt(d))/(2*a); %find root 2

%display the result
disp('determinant :- '); disp(d);
disp('Root 1 :- '); disp(root1);
disp('root 2 :- '); disp(root2);
```

```
determinant :-
    25

Root 1 :-
     1

root 2 :-
    -4
```

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```
%{
Exp 3.2
To find Nonlinear Roots:
f(x) = 2x + e^(-x/2)
find f(0) and roots
%}

f = @(x) 2*x + exp(-x/2);
f0 = f(0) ;
root1 = fzero (f , -1);
root2 = fzero (f , -5);
fprintf('f(0) =%0.6f\n',f0);
fprintf('root1 =%0.6f\n',root1);
fprintf('root2 =%0.6f\n',root2);

f = @(x) 2*x + exp(-x/2);   % Define the function

f0 = f(0);                  % Evaluate f(0)

% Find roots near -1 and -5
root1 = fzero(f, -1);
root2 = fzero(f, -5);

% Display results
fprintf('f(0) = %.6f\n', f0);
fprintf('root1 = %.6f\n', root1);
fprintf('root2 = %.6f\n', root2);
```

```
f(0) =1.000000
root1 =-0.714806
root2 =-4.306585
f(0) = 1.000000
root1 = -0.714806
root2 = -4.306585
```

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```
%{
Exp 3.3
To find Hypotenuse of a
Right triangle with
sides 3, 4
%}

a = 3; b = 4;
c = sqrt(a^2 + b^2);
disp('Hypotenuse = '); disp(c);
```

```
Hypotenuse =
     5
```

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```
%{
Exp 3.4
find the surface area and volume of a right angle cone
having a height of 10 cm and radius of 3 cm
with user input and output
%}

height = 10 ;
radius = 3 ;

curved_area = pi*radius*sqrt(height*height + radius*radius);
total_area = curved_area + pi*radius*radius;
vol = (pi*height*(radius^2))/3;

disp(' curved surface area = ');disp(curved_area);
disp('total area = '); disp(total_area);
disp('volume = ');disp(vol);
```

```
 curved surface area =
   98.3976

total area =
  126.6719

volume =
   94.2478
```

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```
%{
Exp 3.5
Find principle amount compoundly
at interest of 6.1% , principle amount = 100
time = 3 years

Find also SI and CI and their difference
%}

principal = 100; % Principal amount
rate = 6.1; % Interest rate
time = 3; % Time in years

SI = (principal * rate * time) / 100; %Simple Interest
CI = principal * (1 + rate / 100)^time - principal; %Compound Interest
difference = CI - SI; %difference between CI and SI
total_amount = principal + CI;

disp('Total amount = '); disp(total_amount);
disp('simple interest = '); disp(SI);
disp('compund interest = '); disp(CI);
disp('difference between ci and si = '); disp(difference);
```

```
Total amount =
  119.4390

simple interest =
   18.3000

compund interest =
   19.4390

difference between ci and si =
    1.1390
```

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```
%{
Exp 3.6
3x+4y+z=2
4x+y-z=3
y+3z=0
%}

A = [3,4,1;4,1,-1;0,1,3];
B = [2;3;0];

X = (inv(A))*B;
X
```

```
X =

    0.7812
   -0.0938
    0.0312
```

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```
%{
Exp 3.7
Find the surface area and volume of a cone having
a height of 10cm and radius of 5 cm
%}

height = 10;
radius = 5;

curved_area = pi*radius*sqrt(height*height + radius*radius);
total_area = curved_area + pi*radius*radius;
vol = (pi*height*(radius^2))/3;

disp(' curved surface area = ');disp(curved_area);
disp('total area = '); disp(total_area);
disp('volume = ');disp(vol);
```

```
 curved surface area =
  175.6204

total area =
  254.1602

volume =
  261.7994
```

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```
% Exp 4.1
% Sum of first 20 natural numbers number
a = 0;
for i = 1:1:20
    a = a + i;
end
disp(a);
```

```
   210
```

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```
% Exp 4.2
% Find the factorial of 8

% Method 1 using factorial command
a = factorial(8);
disp(a); %display

% Method 2 using a for loop
b = 1;
for i = 1:1:8
    b = b*i;
end

%display
disp(b);
```

```
       40320

       40320
```

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```
%{
Exp 4.3
find how many years are required to make
a sum of 10 thousands to 20 thousands with
rate of interest of 8% compounded annualy
%}

principle = 10000;
total_amount = 20000;
rate = 8;

%{
We know: total_amount = principle * (1 + rate/100)^time
log(total_amount) = log(principle) + time * log(1 + rate/100)
time = (log(total_amount) - log(principle)) / log(1 + rate/100)
%}

time = (log(total_amount) - log(principle)) / log(1 + rate / 100);
disp(time);
```

```
    9.0065
```

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```
% exp 4.4
% Find the roots using cramer's rule

%{
3x+4y-7z=0
4x+9y+z=8
-x-y+z=-4
%}

D = [3,4,-7;4,9,1;-1,-1,1];
disp('rank of D = '); disp(rank(D));

Dx = [0,4,-7;8,9,1;-4,-1,1]; %replaced 1st column with b
disp('rank of Dx = ');disp(rank(Dx));
x = det(Dx)/det(D);
disp('x = '); disp(x);

Dy = [3,0,-7;4,8,1;-1,-4,1]; %replace 2nd column with b
disp('rank of Dy = '); disp(rank(Dy));
y = det(Dy)/det(D);
disp('y = '); disp(y);

Dz = [3,4,0;4,9,8;-1,-1,-4]; %replace 3rd column with b
disp('rank of Dz = '); disp(rank(Dz));
z = det(Dz)/det(D);
disp('z = '); disp(z);
```

```
rank of D =
     3

rank of Dx =
     3

x =
    9.7600

rank of Dy =
     3

y =
   -3.6800

rank of Dz =
     3

z =
    2.0800
```

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```
%{
Exp 4.5
find a grade of a student who scored 81 marks
if A(>=90) and B(<90 and>=80) and C(>=70 and <80)
and D(>=60 and <70) else F.
%}

mark = 81; %define=d the marks

% Determine the grade based on the marks
if mark >= 90
    grade = 'A';
elseif mark >= 80
    grade = 'B';
elseif mark >= 70
    grade = 'C';
elseif mark >= 60
    grade = 'D';
else
    grade = 'F';
end

%display the grade
disp('garde = '); disp(grade);
```

```
garde =
B
```

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```
% Exp 4.6
% Find a number is even or odd

number = 13;

%check number is even or odd
if mod(number , 2) == 0
    disp('Even')
else
    disp('Odd')
end
```

```
Odd
```

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```
% Exp 4.7
% Check if a number is prime or not

number = 57;

count = 0;
for i = 1:1:number %for loop from 1 to n
    if mod(number , i) == 0 %check if that number is divisable bw 1 to n
        count = count + 1; %adds 1 if fully divisable
    end
end

if count == 2
    disp('prime')
else
    disp('not prime')
end
```

```
not prime
```

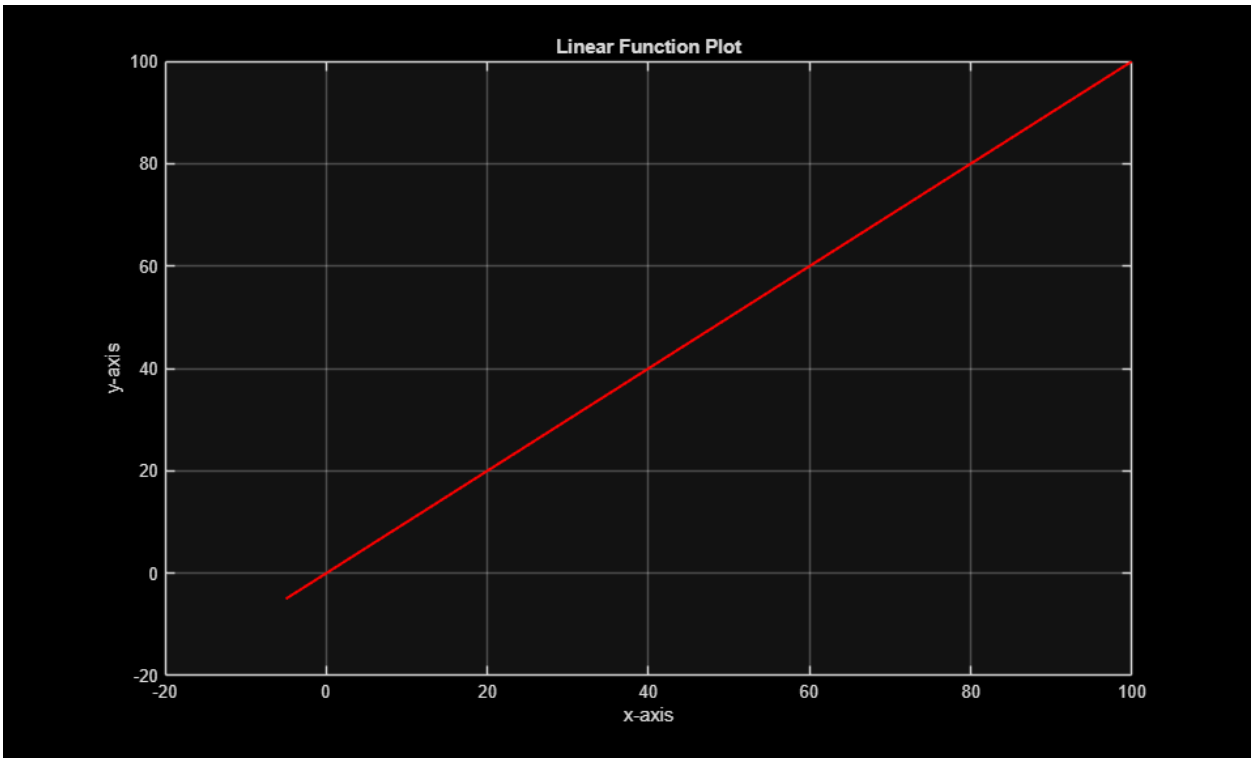
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Q1) Plotting a Straight Line Graph: y = m*x + c

```
x = -5:1:100;                                   % Define x range
m = 1;  % slope
c = 0; % intercept

y = m*x + c;                                    % Linear equation

plot(x, y, 'r', 'LineWidth', 1.5);
xlabel('x-axis');
ylabel('y-axis');
title('Linear Function Plot');
grid on;
```

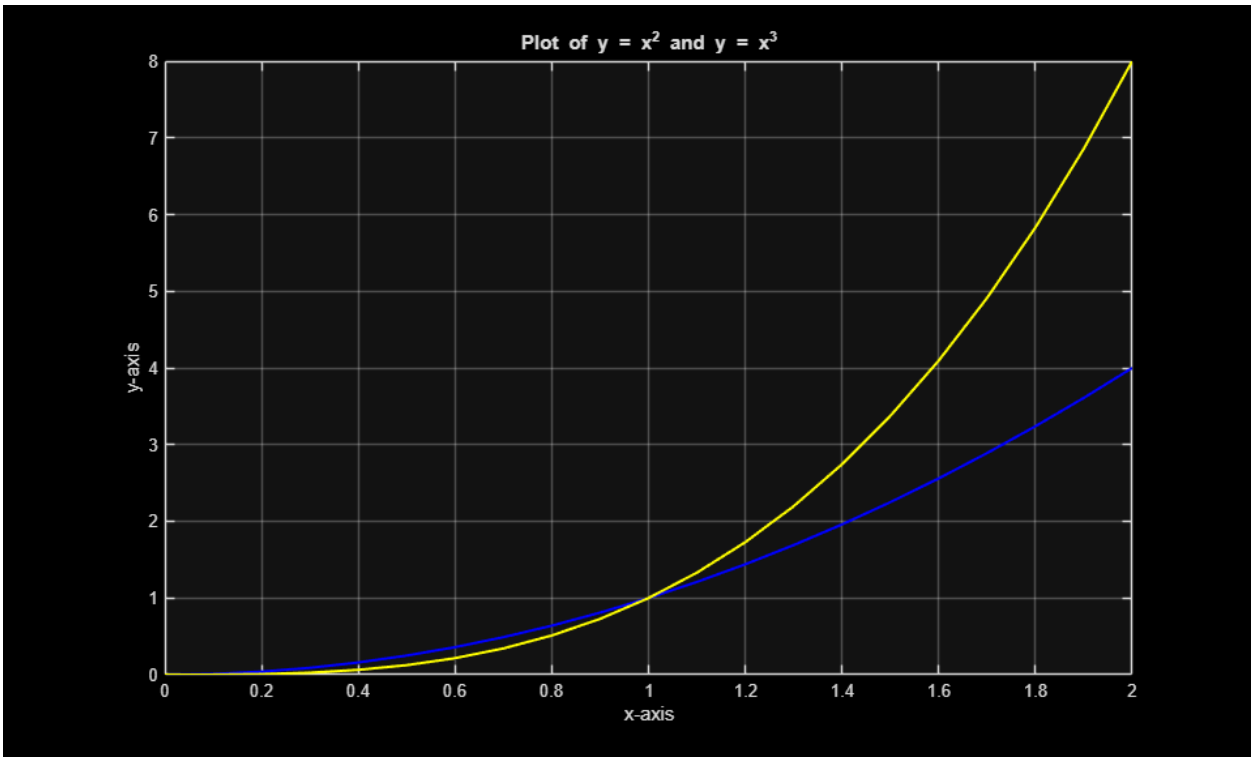


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Q2) Plotting y = x^n Function

```
x = 0:0.1:2;      % Define x range
y1 = x.^2;        % y = x^2
y2 = x.^3;        % y = x^3 (added for better demonstration)

figure;
plot(x, y1, 'b', 'LineWidth', 1.5); hold on;
plot(x, y2, 'y', 'LineWidth', 1.5);
xlabel('x-axis');
ylabel('y-axis');
title('Plot of y = x^2 and y = x^3');
grid on;
```

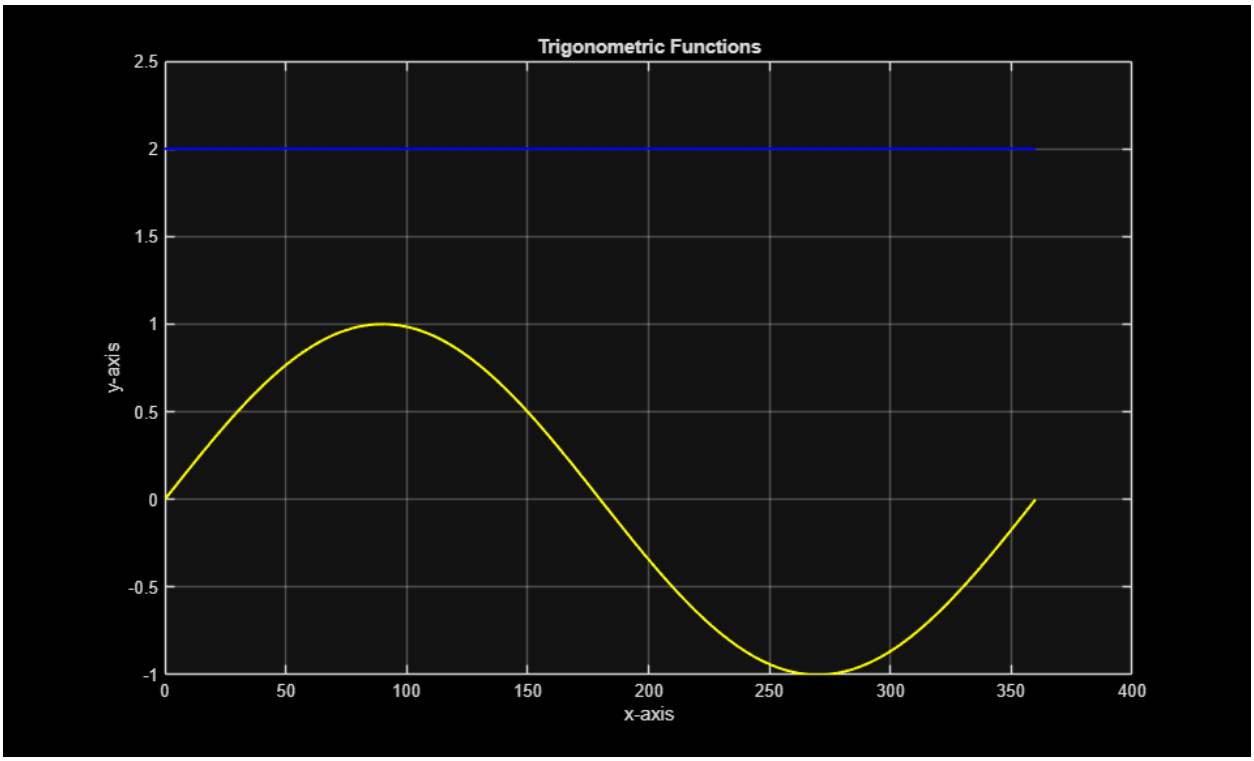


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Q3) Trigonometric Function Plot

```
x = 0:360;              % Degrees
y1 = sind(x);           % sin(x)
y2 = 2*((sind(x)).^2 + (cosd(x)).^2);   % Derived function

figure;
plot(x, y1, 'y', 'LineWidth', 1.5); hold on;
plot(x, y2, 'b', 'LineWidth', 1.5);
xlabel('x-axis');
ylabel('y-axis');
title('Trigonometric Functions');
grid on;
```

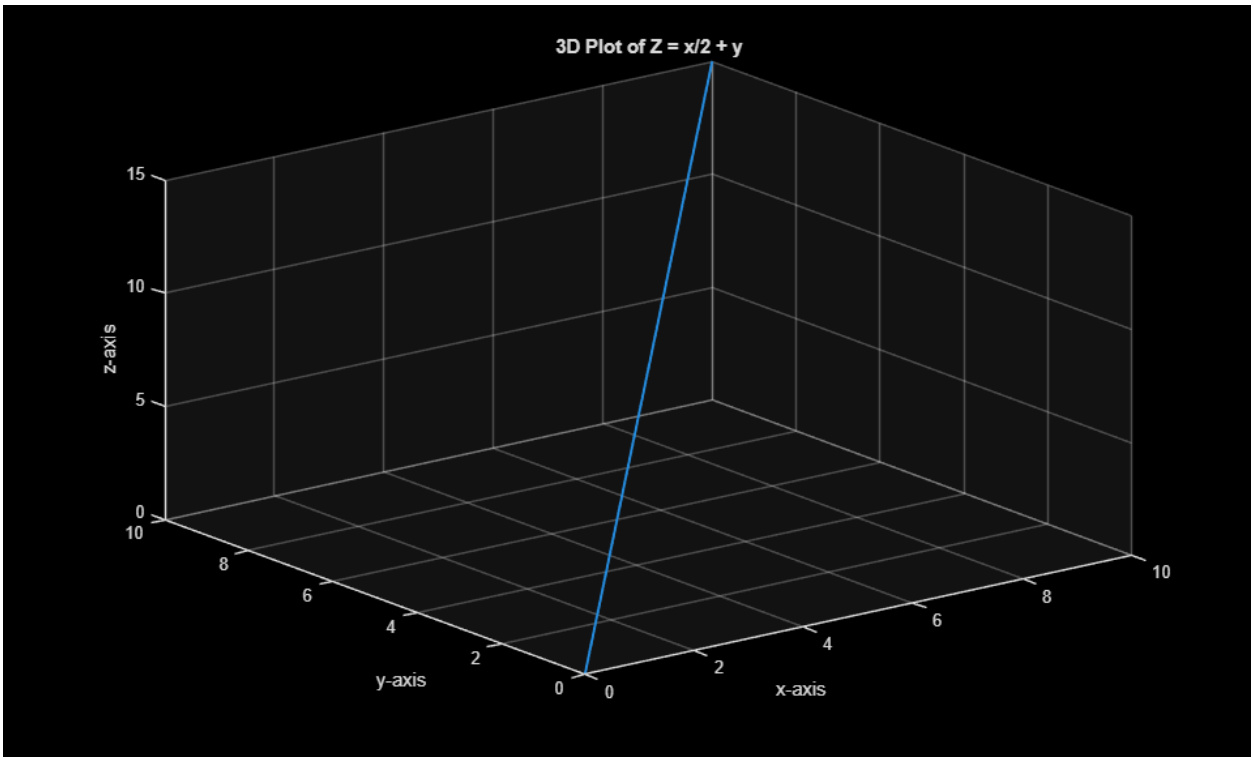


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Q4) 3D Plot of Z = x/2 + y

```
x = 0:0.1:10;
y = x;                  % Use same range for y
z = x/2 + y;            % Equation for z

figure;
plot3(x, y, z, 'LineWidth', 1.5);
xlabel('x-axis');
ylabel('y-axis');
zlabel('z-axis');
title('3D Plot of Z = x/2 + y');
grid on;
```

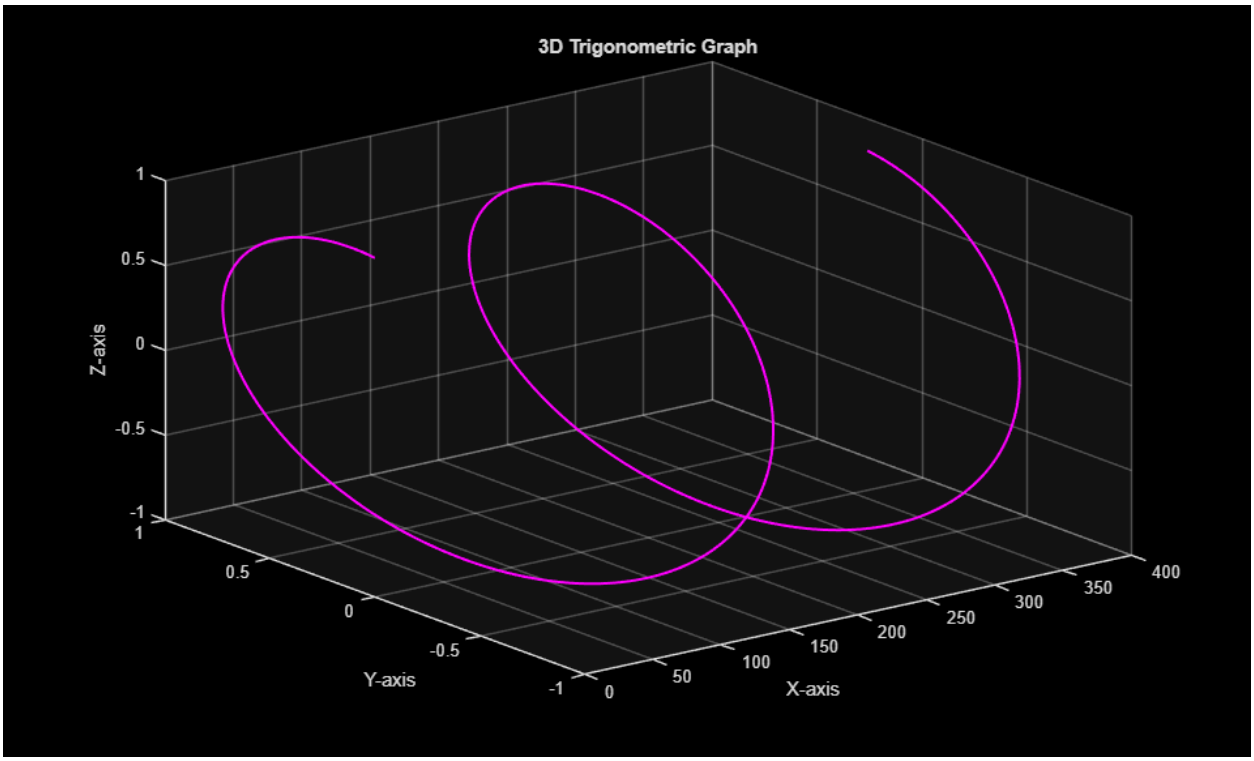


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Q5) 3D Trigonometric Function

```
x = 0:360;
y = sind(2*x);
z = cosd(2*x);

figure;
plot3(x, y, z, 'm', 'LineWidth', 1.5);
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Z-axis');
title('3D Trigonometric Graph');
grid on;
```

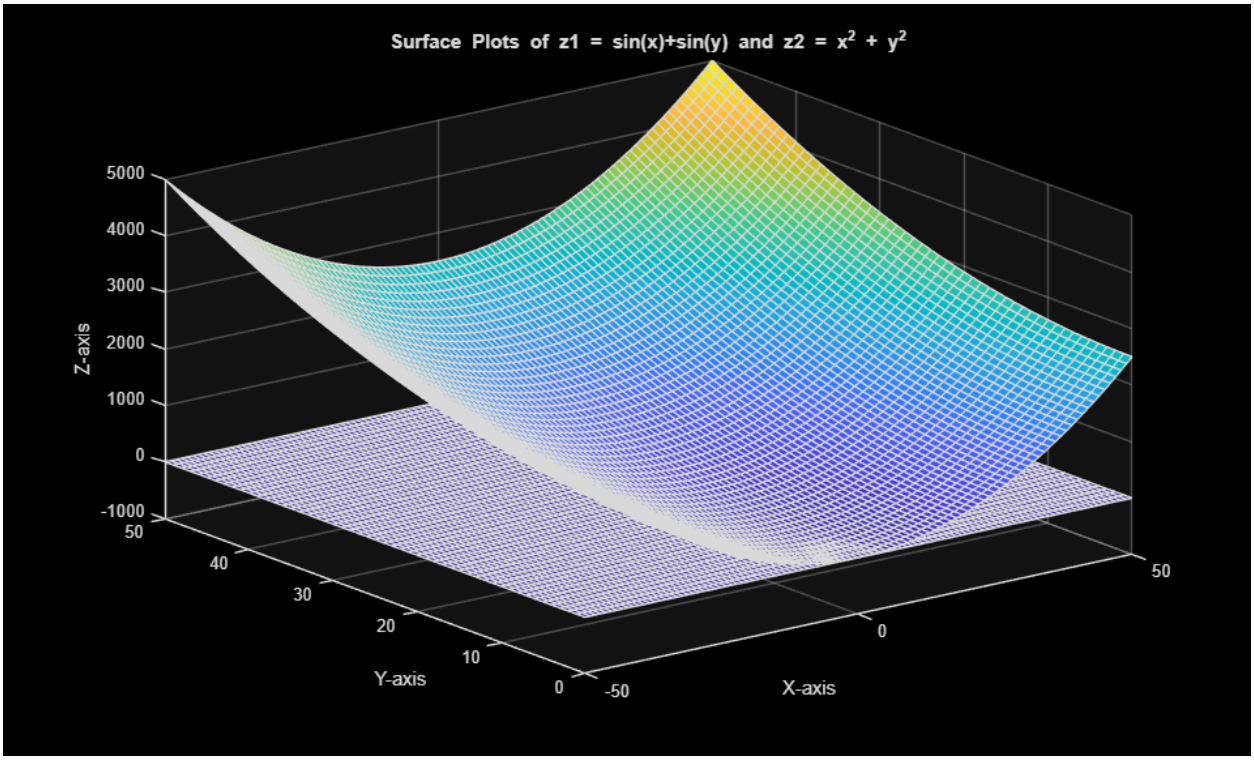


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Q6) Using Meshgrid Function

```
x = -50:1:50;
y = 0:1:50;
[X, Y] = meshgrid(x, y);        % Create grid
Z1 = sin(X) + sin(Y);           % First surface
Z2 = X.^2 + Y.^2;               % Second surface

figure;
surf(X, Y, Z1);
hold on;
surf(X, Y, Z2);
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Z-axis');
title('Surface Plots of z1 = sin(x)+sin(y) and z2 = x^2 + y^2');
grid on;
```



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